



DUBLIN CITY UNIVERSITY

SEMESTER ONE EXAMINATIONS 2013

MODULE: (Title & Code)	CA208 Logic
COURSE:	BSc. in Computer Applications (SE Stream)
YEAR:	2
EXAMINERS: (Including Telephone Nos.)	Dr. J. Power, Dr. M. Manzke, Dr. D. Sinclair, Ext no. 5510.
TIME ALLOWED:	2 hours
INSTRUCTIONS:	Answer 10 Questions. Answer all 8 Questions in Section A. Answer any 2 questions from Section B. All questions carry equal marks.

Please do not turn over this page until instructed to do so

The use of programmable or text storing calculators is expressly forbidden.
Please note that where a candidate answers more than the required number of questions, the examiner will mark all questions attempted and then select the highest scoring ones

Section A

QUESTION 1

[TOTAL MARKS: 10]

[10 Marks]

Explain the following:

- satisfiability
- validity
- refutation procedure

[End of Question 1]

QUESTION 2

[TOTAL MARKS: 10]

[10 Marks]

Show that

$$(A \vee B) \rightarrow C \equiv (A \rightarrow C) \wedge (B \rightarrow C)$$

[End of Question 2]

QUESTION 3

[TOTAL MARKS: 10]

[10 Marks]

Using semantic tableaux, prove the following propositional formula is valid.

$$(p \rightarrow q) \rightarrow ((p \rightarrow \neg q) \rightarrow \neg p)$$

[End of Question 3]

QUESTION 4

[TOTAL MARKS: 10]

[10 Marks]

Using the Gentzen System $\mathcal{G}$ prove

$$\vdash (A \rightarrow B) \rightarrow (\neg B \rightarrow \neg A)$$

[End of Question 4]

QUESTION 5**[TOTAL MARKS: 10]****[10 Marks]**

Using the Hilbert System $\mathcal{H}$ and the following derived rules and theorem,

$$\text{Deduction: } \frac{U \cup \{A\} \vdash B}{U \vdash A \rightarrow B}$$

$$\text{Contrapositive: } \frac{U \vdash A \rightarrow B}{U \vdash \neg B \rightarrow \neg A}$$

$$\text{Double Negation: } \frac{U \vdash \neg \neg A}{U \vdash A}$$

$$\text{Theorem: } \vdash (\neg A \rightarrow \neg \neg A) \rightarrow \neg \neg A$$

prove

$$\vdash (\neg A \rightarrow A) \rightarrow A$$

[End of Question 5]

QUESTION 6**[TOTAL MARKS: 10]****[10 Marks]**

Convert the following propositional formula into Conjunctive Normal Form and using resolution show that this formula is unsatisfiable.

$$(p \vee q) \wedge (q \rightarrow r) \wedge \neg r \wedge (p \rightarrow s) \wedge (s \rightarrow \neg t) \wedge t$$

[End of Question 6]

QUESTION 7**[TOTAL MARKS: 10]****[10 Marks]**

Using Hilbert System $\mathcal{H}$ and the following derived rules

$$\text{Deduction: } \frac{U \cup \{A\} \vdash B}{U \vdash A \rightarrow B}$$

$$\text{Contrapositive: } \frac{U \vdash A \rightarrow B}{U \vdash \neg B \rightarrow \neg A}$$

prove

$$\vdash \forall x.(p(x) \rightarrow q) \rightarrow \forall x.(\neg q \rightarrow \neg p(x))$$

[End of Question 7]

QUESTION 8**[TOTAL MARKS: 10]****[10 Marks]**

Using Gentzen System $\mathcal{G}$ prove

$$\vdash \forall x.(p(x) \rightarrow q(x)) \rightarrow (\exists x.p(x) \rightarrow \exists x.q(x))$$

[End of Question 8]

Section B**QUESTION 9****[TOTAL MARKS: 10]****[10 Marks]**

Given the following Prolog relations:

- `parent(X,Y)` is true if X is the parent of Y,
- `male(X)` is true if X is male, and
- `female(X)` is true if X is female,

write the following Prolog relations:

- `brother(X,Y)` is true if X is the brother of Y
- `uncle(X,Y)` is true if X is the uncle of Y

[End of Question 9]

QUESTION 10**[TOTAL MARKS: 10]****[10 Marks]**

Write a Prolog relation `nodups(X,Y)` that is true if the elements of list Y are the elements of list X without any duplicate elements.

For example, `nodups([1,3,2,3,4,1,5],[2,3,4,1,5])` is true.

[End of Question 10]

QUESTION 11**[TOTAL MARKS: 10]****[10 Marks]**

Describe what *negation as failure* means in Prolog. Why is negation problematical in Prolog? Illustrate your answer with examples.

[End of Question 11]

Appendix

Semantic Tableaux

α	α_1	α_2	β	β_1	β_2
$\neg\neg A_1$	A_1		$\neg(B_1 \wedge B_2)$	$\neg B_1$	$\neg B_2$
$A_1 \wedge A_2$	A_1	A_2	$B_1 \vee B_2$	B_1	B_2
$\neg(A_1 \vee A_2)$	$\neg A_1$	$\neg A_2$	$B_1 \rightarrow B_2$	$\neg B_1$	B_2
$\neg(A_1 \rightarrow A_2)$	A_1	$\neg A_2$	$B_1 \uparrow B_2$	$\neg B_1$	$\neg B_2$
$\neg(A_1 \uparrow A_2)$	A_1	A_2	$\neg(B_1 \downarrow B_2)$	B_1	B_2
$A_1 \downarrow A_2$	$\neg A_1$	$\neg A_2$	$\neg(B_1 \leftrightarrow B_2)$	$\neg(B_1 \rightarrow B_2)$	$\neg(B_2 \rightarrow B_1)$
$A_1 \leftrightarrow A_2$	$A_1 \rightarrow A_2$	$A_2 \rightarrow A_1$	$B_1 \oplus B_2$	$\neg(B_1 \rightarrow B_2)$	$\neg(B_2 \rightarrow B_1)$
$\neg(A_1 \oplus A_2)$	$A_1 \rightarrow A_2$	$A_2 \rightarrow A_1$			

γ	$\gamma(a)$	δ	$\delta(a)$
$\forall x.A(x)$	$A(a)$	$\exists x.A(x)$	$A(a)$
$\neg\exists x.A(x)$	$\neg A(a)$	$\neg\forall x.A(x)$	$\neg A(a)$

Gentzen Logic

$$\frac{\vdash U'_1 \cup \{\alpha_1, \alpha_2\}}{\vdash U'_1 \cup \{\alpha\}} \quad \frac{\vdash U'_1 \cup \{\beta_1\} \quad \vdash U'_2 \cup \{\beta_2\}}{\vdash U'_1 \cup U'_2 \cup \{\beta\}}$$

$$\frac{\vdash U'_1 \cup \{\gamma, \gamma(a)\}}{\vdash U'_1 \cup \{\gamma\}} \quad \frac{\vdash U'_1 \cup \{\delta(a)\}}{\vdash U'_1 \cup \{\delta\}}$$

$\frac{A}{B}$ is a *sequent* and means that if A is true, then B can be inferred.

α	α_1	α_2	β	β_1	β_2
$\neg\neg A$	A		$B_1 \wedge B_2$	B_1	B_2
$\neg(A_1 \wedge A_2)$	$\neg A_1$	$\neg A_2$	$\neg(B_1 \vee B_2)$	$\neg B_1$	$\neg B_2$
$A_1 \vee A_2$	A_1	A_2	$\neg(B_1 \rightarrow B_2)$	B_1	$\neg B_2$
$A_1 \rightarrow A_2$	$\neg A_1$	A_2	$\neg(B_1 \uparrow B_2)$	B_1	B_2
$A_1 \uparrow A_2$	$\neg A_1$	$\neg A_2$	$B_1 \downarrow B_2$	$\neg B_1$	$\neg B_2$
$\neg(A_1 \downarrow A_2)$	A_1	A_2	$B_1 \leftrightarrow B_2$	$B_1 \rightarrow B_2$	$B_2 \rightarrow B_1$
$\neg(A_1 \leftrightarrow A_2)$	$\neg(A_1 \rightarrow A_2)$	$\neg(A_2 \rightarrow A_1)$	$\neg(B_1 \oplus B_2)$	$B_1 \rightarrow B_2$	$B_2 \rightarrow B_1$
$A_1 \oplus A_2$	$\neg(A_1 \rightarrow A_2)$	$\neg(A_2 \rightarrow A_1)$			

γ	$\gamma(a)$	δ	$\delta(a)$
$\exists x.A(x)$	$A(a)$	$\forall x.A(x)$	$A(a)$
$\neg\forall x.A(x)$	$\neg A(a)$	$\neg\exists x.A(x)$	$\neg A(a)$

Hilbert Logic

The *Hilbert System* $\mathcal{H}$ for propositional logic has 3 axioms,

Axiom 1: $\vdash (A \rightarrow (B \rightarrow A))$

Axiom 2: $\vdash (A \rightarrow (B \rightarrow C)) \rightarrow ((A \rightarrow B) \rightarrow (A \rightarrow C))$

Axiom 3: $\vdash (\neg B \rightarrow \neg A) \rightarrow (A \rightarrow B)$

and only 1 rule of inference.

Modus Ponens:
$$\frac{\vdash A \quad \vdash A \rightarrow B}{\vdash B}$$

The *Hilbert System* $\mathcal{H}$ for first-order logic has 2 additional axioms,

Axiom 4: $\vdash \forall x.A(x) \rightarrow A(a)$ x is a free variable in $A(x)$

Axiom 5: $\vdash \forall x.(A \rightarrow B(x)) \rightarrow (A \rightarrow \forall x.B(x))$ x is a free variable in $B(x)$
 x is not free in $A(x)$

and 1 additional rule of inference.

Generalisation:
$$\frac{\vdash A(a)}{\vdash \forall x.A(a)}$$

[END OF EXAM]